

Lab Worksheet 4: Iteration

CS 101 Algorithmic Problem Solving

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1. Working your way up

In your math class today you were asked to calculate the sum of digits of a particular number, Your job is now to write psuedocode to do that for you so that you can easily complete your math homework assignment.

Constraints

- $1 \leq N \leq 10^8$

Interaction

The input contains a single integer value denoting the number whose digits you have to sum

The output would contain the sum of the digits.

Sample

Input	Output
1829	20
23	5

In the first case, $N = 1829$, therefore the answer would be $1 + 8 + 2 + 9 = 20$

In the second case, $N = 23$, therefore the answer would be $2 + 3 = 5$

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2872	19
98723	29

Pseudocode

```
Input: N
Output:
a=0
b=0
while N>0:
    a=N%10
    N=N//10
    b=b+a
print(b)
```

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your psuedocode in the space below.

```

Input: 2872          while 28>0:(true)
N=2872              a=28%10
a=0                  N=28//10
b=0                  b=9+8
while 2872>0:(true) while 2>0: (true)
    a=2872%10        a=2%10
    N=2872//10        N=2//10
    b=0+2             b=17+2
while 287>0:(true)   while 0>0: (false)
    a=287%10          print(b) (19)
    N=287//10
    b=2+7              OUTPUT: 19

```

2. Party Invites

Your friend is hosting a party at his house. He's prepared X burgers and Y sandwiches for the event. He needs your help in figuring out how many people to invite. Ideally, he wants that everyone gets the same amount of burgers and sandwiches, and no food is leftover.

Given values of X and Y , determine the maximum number of guests your friend can invite.

Constraints

- $X, Y \in \mathbb{N}$
- $1 \leq X, Y \leq 10^5$

Interaction

The input comprises of a single line containing two space-separated integers denoting the values of X and Y respectively.

The output must be a single integer denoting the maximum number of guests your friend can invite.

Sample

Input	Output
12 16	4
9 9	9

In the first case, $(X, Y) = (12, 16)$, the maximum number of guests your friend can invite is 4; each guest can have 3 burgers and 4 sandwiches.

In the second case, $(X, Y) = (9, 9)$, the maximum number of guests your friend can invite is 9; each guest can have 1 burger and 1 sandwich.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
16 17	1
32 52	4
256 128	128

Pseudocode

Input: X, Y

Output:

```
if X<Y:
    for i in range(X+1,1,-1):
        if X%i==0 and Y%i==0:
            print(i)
            break
else:
    for i in range(Y+1,1,-1):
        if X%i==0 and Y%i==0:
            print(i)
            break
```

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: 128, 256

```
if X<Y: (true)
    for i in range(128+1,1,-1):
        if 128%i==0 and 256%i==0: (true)
            print(i) (i=128)
            break
else: (false)
    for i in range(Y+1,1,-1):
        if X%i==0 and Y%i==0:
            print(i)
            break
```

Output= 128

3. Magic Potions

Aimen plays a game in which the character competes in a hurdle race. Hurdles are of varying heights, and the characters have a maximum height h they can jump. There is a magic potion they can take that will increase their maximum jump height by 1 unit for each dose.

Given the number of jumps n and maximum height h , figure out how many doses of the potion must Aimen's character take to be able to jump all of the hurdles. If Aimen can already clear all of the hurdles without taking the potion, return 0.

Constraints

- $1 \leq n, h \leq 25$

Interaction

The input comprises a line containing 2 space-separated integers denoting the values of n and h respectively. Followed by a line containing n integers separated by space, each denoting the height of a hurdle.

The output must contain a single number denoting the total doses of magic potions needed.

Sample

Input	Output
4 3	0
1 2 3 2	
5 7	3
2 7 8 10 3	

In the first case, all hurdles are lower than the maximum height the character can jump. Hence, no magic potion needed.

In the second case, the tallest hurdle is 10 hence 3 doses of the magic potion are needed. If they are taken, the character will be able to jump the hurdles 8 and 10 which are greater than the maximum height.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4 3	6
9 6 1 2	
6 8	4
1 3 8 9 10 12	

Pseudocode

Input: n,h

Output:

a=0

for i in range(n):

 n1=int(input("enter height of hurdle: "))

 if n1>a:

 a=n1

print(a-h)

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: n=4, h=3	for i in range(n):
Output:	n1=1
a=0	if n1>a: (false)
for i in range(n):	a=9
n1=9	for i in range(n):
if n1>a: (true)	n1=2
a=9	if n1>a: (false)
for i in range(n):	a=9
n1=6	print(a-h) (9-3)
if n1>a: (false)	
a=9	Output: 6

4. Password

You've made a website that hosts information about upcoming events taking place at your university. You're working on the backend of the login system and have decided that the user passwords must be very difficult to crack so that nobody can impersonate others and misuse your website. You've come up with a set of parameters that a password must satisfy in order to be accepted. These are:

- The password must contain a lowercase character.
- The password must contain an uppercase character.
- The password must contain a number.
- The password must not contain any symbols or spaces.

Given a string S , determine if it is an acceptable password or not.

Constraints

- $1 \leq \text{len}(S) \leq 20$

Interaction

The input comprises of a single line containing a string S denoting the password.

The output must be "YES" if the string is an acceptable password, and "NO" otherwise.

Sample

Input	Output
"seVenteen17"	"YES"
"H@B1Buni"	"NO"

In the first case, $S = \text{"seVenteen17"}$ which satisfies all the given parameters and is an acceptable password.

In the second case, $S = \text{"H@B1Buni"}$ which contains a special character and is not an acceptable password.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
"pass_Word23"	"NO"
"29thFeb2004"	"YES"
"14th Aug"	"NO"

Pseudocode

Input: S

Output:

l=False

t=False

for i in S:

if i.isalpha():

if S.isupper():

t=False

break

elif S.islower():

t=False

break

else:

t=True

elif i.isnumeric():

l=True

if S.isnumeric():

break

else:

l=False

break

if l==True and t==True:

print("YES")

else:

print("NO")

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

Input: pass_Word23

l= False

t= False

(CHECKING FOR P)

for i.isalpha():

if S.isupper(): (t=false)

elif S.islower(): (t=false)

else: (working condition)

t=True

(CHECKING FOR A)

for i.isalpha():

if S.isupper(): (t=false)

elif S.islower(): (t=false)

else: (working condition)

t=True

(CHECKING FOR S)

for i.isalpha():

if S.isupper(): (t=false)

elif S.islower(): (t=false)

else: (working condition)

t=True

(CHECKING FOR S)

for i.isalpha():

if S.isupper(): (t=false)

elif S.islower(): (t=false)

else: (working condition)

t=True

(CHECKING FOR _)

for i.isalpha(): (false)

if S.isupper(): (t=false)

elif S.islower(): (t=false)

else:

t=True

NOW THE t = FALSE

elif i.isnumeric(): (False)

l=True

if S.isnumeric(): (False)

break

else: (WORKING CONDITION)

l=False

break

if l==True and t==True: (FALSE)

print("YES")

else: (WORKING CONDITION)

print("NO")

Output: "NO"

5. War in the Milkyway

Ali is playing a game 'War in the Milkyway'. The Milkyway is a galaxy with N planets, two of which have power sources namely $R1$ and $R2$. Planets are numbered consecutively from 0 and each planet is at a distance of 1 km to the next planet. It is not circular, so the first planet doesn't connect with the last planet.

Ali needs to determine the maximum distance from any planet to its nearest power source, in order to plan efficiently. Help him with this task.

Hint: using `min()`, `max()` and `abs()` may help.

Constraints

- $0 \leq R1, R2 \leq 100$
- $1 \leq N \leq 100$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of N , $R1$ and $R2$ respectively.

The output must contain a single number denoting the maximum distance.

Sample

Input	Output
5 0 4	2
3 6 7	0

In the first case, there are a total of 5 planets. Planet 0 and 4 have the distance of 0 to the nearest power source. Planet 1 and 3 have power source 0 and 4 respectively at a distance of 1 km. Planet 2 is 2 km away from both the power sources hence, the maximum distance for all 5 planets is 2 km.

In the second case, there are 3 planets but stations are said to be at planets numbered 6 and 7 which do not exist, hence the output is 0.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
7 0 4	2
4 9 6	0
11 2 10	4

Pseudocode

Input: N , $R1$, $R2$

Output:

answer=0

if $R1 > N$ or $R2 > N$:

 print(0)

for i in range(N):

$a1 = \text{abs}(R1 - i)$

$a2 = \text{abs}(R2 - i)$

 if $a1 \leq a2$ and $a1 > \text{answer}$:

 answer= $a1$

 elif $a2 \leq a1$ and $a2 > \text{answer}$:

 answer= $a2$

print(answer)

Dry Run

Using any one of the inputs provided in the Exercise section above, dry run your psuedocode in the space below.

```

Input: 4,9,6
answer=0
if 9>4 OR 6>4: (true)
    print(0)
for i in range(4):
    a1=abs(9-i)
    a2=abs(6-i)
    if a1<=a2 and a1>answer: (false)
        answer=a1
    elif a2<=a1 and a2>answer: (false)
        answer=a2
print(answer)

```

Output: 0

6. Iterative exponent

You have studied exponents both early in this class and in secondary school, and your friends challenge you to find the exponent using multiplications only. Given m and n two integers your job is to find the exponent is the form m^n

Constraints

- $1 \leq m, n \leq 10^3$

Interaction

The input contains two space-separated integers m and n .

Output would contain a single integer which is m^n

Sample

Input	Output
3,4	81
0,2	0

In the first case, $m, n = 3, 4$, $3^4 = 81 = 3 * 3 * 3 * 3$

In the second case, $m, n = 0, 2$, $0^2 = 0 = 0 * 0$

Proposed Solution

```

1     product = m
2     if m == 0:
3         print(0)
4
5     elif n == 0:
6         print(0)
7
8     else:
9         for i in range(0,n-1):
10            product = product**m
11
12        print(product)

```

Dry Run

Using the inputs provided in the Sample section above, dry run the proposed code solution below.

```
Input: 0,2
if m==0: (true)
    print(0)
elif n==0: (false)
    print(0)
else:
for i in range(0,n-1):
    product=product**m
print(product)
```

Output: 0

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and the errors in each line.

in line 6, it should be print 1, in line 9 the range(n) should be written, in line 10 the " * " should be used instead of " ** "

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

Input: m, n

```
Output:
product=1
if m==0:
    print(0)
elif n==0:
    print(1)
else:
    for i in range(n):
        product=product*m

print(product)
```

7. Introduction to Hackerrank

The guide to login/sign-up on Hackerrank tests is available here.

Problems are available on this link as a short introduction to programming:

<https://hr.gs/cs101-lw4-fall24>

Rough Work